

Discrete System Property Analysis

DSP - Lab 01

Types of discrete systems

- **causal** – only depends on previous input values:
$$y[n] = f(x[n], x[n-1], x[n-2], \dots)$$
- **memory-less** – only depends on the current input value:
$$y[n] = f(x[n])$$
- **delay** – shifts a sequence in time:
$$y[n] = x[n-d]$$
- **time-invariant** $\mathcal{H}$ if
$$y[n] = \mathcal{H}(x[n]) \Leftrightarrow y[n-d] = \mathcal{H}(x[n-d])$$
- **linear** $\mathcal{H}$ if
$$\mathcal{H}(a_1x_1[n] + a_2x_2[n]) = a_1\mathcal{H}(x_1[n]) + a_2\mathcal{H}(x_2[n])$$
- **stable** if
$$|x[n]| \leq M_x < \infty \Rightarrow |y[n]| \leq M_y < \infty$$

What type of discrete system ((non-)linear, (non-)time-invariant, (non-)causal, memory-less, etc.) is

1. $y[n] = |x[n]|$

Causal: Depends only on $x[n]$. ✓

Memory-less: Uses only the present input. ✓

Delay: A delay system has form $y[n] = x[n-d]$. Not satisfied. ✗

Time Invariance

$$\left. \begin{aligned} \mathcal{H}(x[n-d]) &= |x[n-d]| \\ y[n-d] &= |x[n-d]| \end{aligned} \right\} \Rightarrow \mathcal{H}(x[n-d]) = y[n-d] \quad \checkmark$$

Linearity

1. Homogeneity

$$\left. \begin{aligned} \mathcal{H}(ax[n]) &= |ax[n]| = |a||x[n]| \\ a\mathcal{H}(x[n]) &= a|x[n]| \end{aligned} \right\} \Rightarrow \mathcal{H}(ax[n]) \neq a\mathcal{H}(x[n]), \text{ if } a < 0 \quad \text{✗}$$

2. Additivity

$$\mathcal{H}(x_1[n] + x_2[n]) = |x_1[n] + x_2[n]| \leq |x_1[n]| + |x_2[n]| = \mathcal{H}(x_1[n]) + \mathcal{H}(x_2[n]) \quad \text{✗}$$

BIBO Stability

If $|x[n]| \leq M$ then $|y[n]| = |x[n]| \leq M$. ✓

	Causal	Memory-less	Delayed	Time invariant	Linear	Stable
Summary	✓	✓	✗	✓	✗	✓

$$\textcircled{2} \quad y[n] = -x[n-1] + 2x[n] - x[n+1]$$

Causal: Term $x[n+1]$ uses future input. ✗

Memory-less: Uses multiple samples. ✗

Delay: Not a pure shift. ✗

Time Invariance

$$\left. \begin{aligned} \mathcal{H}(x[n-d]) &= -x[n-1-d] + 2x[n-d] - x[n+1-d] \\ y[n-d] &= -x[n-d-1] + 2x[n-d] - x[n-d+1] \end{aligned} \right\} \Rightarrow \mathcal{H}(x[n-d]) = y[n-d] \quad \checkmark$$

Linearity

1. Homogeneity

$$\begin{aligned} \mathcal{H}(ax[n]) &= -ax[n-1] + 2ax[n] - ax[n+1] = \\ &= a(-x[n-1] + 2x[n] - x[n+1]) = a\mathcal{H}(x[n]) \quad \checkmark \end{aligned}$$

2. Additivity: Let the input be $x[n] = x_1[n] + x_2[n]$

$$\begin{aligned} \mathcal{H}(x_1[n] + x_2[n]) &= -(x_1[n-1] + x_2[n-1]) + 2(x_1[n] + x_2[n]) - (x_1[n+1] + x_2[n+1]) = \\ &= (-x_1[n-1] + 2x_1[n] - x_1[n+1]) + (-x_2[n-1] + 2x_2[n] - x_2[n+1]) = \\ &= \mathcal{H}(x_1[n]) + \mathcal{H}(x_2[n]) \quad \checkmark \end{aligned}$$

BIBO Stability If $|x[n]| \leq M$, then

$$|y[n]| = |-x[n-1] + 2x[n] - x[n+1]| \leq |x[n-1]| + 2|x[n]| + |x[n+1]| \leq 4M \quad \checkmark$$

	Causal	Memory-less	Delayed	Time invariant	Linear	Stable
Summary	✗	✗	✗	✓	✓	✓

$$\textcircled{3} \quad y[n] = \sum_{k=0}^8 x[n-k]$$

Causal: Uses present and past inputs. ✓

Memory-less: Depends on multiple samples. ✗

Delay: Not a pure delay. ✗

Time Invariance

$$\left. \begin{aligned} \mathcal{H}(x[n-d]) &= \sum_{k=0}^8 x[n-k-d] \\ y[n-d] &= \sum_{k=0}^8 x[n-d-k] \end{aligned} \right\} \Rightarrow \mathcal{H}(x[n-d]) = y[n-d] \quad \checkmark$$

Linearity

1. Homogeneity

$$\begin{aligned} \mathcal{H}(ax[n]) &= \sum_{k=0}^8 ax[n-k] = \\ &= a \sum_{k=0}^8 x[n-k] = a^9 \mathcal{H}(x[n]) \quad \times \end{aligned}$$

2. Additivity: Let the input be $x[n] = x_1[n] + x_2[n]$

$$\begin{aligned} \mathcal{H}(x_1[n] + x_2[n]) &= \sum_{k=0}^8 (x_1[n-k] + x_2[n-k]) = \\ &= \sum_{k=0}^8 x_1[n-k] + \sum_{k=0}^8 x_2[n-k] = \mathcal{H}(x_1[n]) + \mathcal{H}(x_2[n]) \quad \checkmark \end{aligned}$$

BIBO Stability If $|x[n]| \leq M$, then

$$|y[n]| = \left| \sum_{k=0}^8 x[n-k] \right| \leq \sum_{k=0}^8 |x[n-k]| \leq 9M \quad \checkmark$$

Summary

Causal	Memory-less	Delayed	Time invariant	Linear	Stable
$\checkmark$	$\times$	$\times$	$\checkmark$	$\times$	$\checkmark$

4. $y[n] = \frac{1}{2}(x[2n] + x[2n+1])$

Causal: Future inputs are required. $\times$

Memory-less: Uses multiple samples. $\times$

Delay: Not a pure delay. $\times$

Time Invariance

$$\left. \begin{aligned} \mathcal{H}(x[n-d]) &= \frac{1}{2}(x[2n-d] + x[2n+1-d]) \\ y[n-d] &= \frac{1}{2}(x[2n-2d] + x[2n-2d+1]) \end{aligned} \right\} \Rightarrow \mathcal{H}(x[n-d]) \neq y[n-d] \quad \times$$

Linearity

1. Homogeneity

$$\begin{aligned} \mathcal{H}(ax[n]) &= \frac{1}{2}(ax[2n] + ax[2n+1]) = \\ &= a \frac{1}{2}(x[2n] + x[2n+1]) = a \mathcal{H}(x[n]) \quad \checkmark \end{aligned}$$

2. Additivity: Let the input be $x[n] = x_1[n] + x_2[n]$

$$\begin{aligned}\mathcal{H}(x_1[n] + x_2[n]) &= \frac{1}{2}[(x_1[2n] + x_2[2n]) + (x_1[2n+1] + x_2[2n+1])] = \\ &= \frac{1}{2}(x_1[2n] + x_1[2n+1]) + \frac{1}{2}(x_2[2n] + x_2[2n+1]) = \mathcal{H}(x_1[n]) + \mathcal{H}(x_2[n]) \quad \checkmark\end{aligned}$$

BIBO Stability If $|x[n]| \leq M$, then

$$|y[n]| = \left| \frac{1}{2}(x[2n] + x[2n+1]) \right| \leq \frac{1}{2}(|x[2n]| + |x[2n+1]|) \leq M \quad \checkmark$$

	Causal	Memory-less	Delayed	Time invariant	Linear	Stable
Summary	✗	✗	✗	✗	✓	✓

5. $y[n] = \frac{3x[n-1] + x[n-2]}{x[n-3]}$

Causal: Uses past inputs. ✓

Memory-less: Uses multiple inputs. ✗

Delay: Not a simple shift. ✗

Time Invariance: Indices shift consistently. ✓

Linearity

1. Homogeneity

$$\mathcal{H}(ax[n]) = \frac{3ax[n-1] + ax[n-2]}{ax[n-3]} = \frac{3x[n-1] + x[n-2]}{x[n-3]} \neq a\mathcal{H}(x[n]) \quad \text{✗}$$

2. Additivity: Let the input be $x[n] = x_1[n] + x_2[n]$

$$\begin{aligned}\mathcal{H}(x_1[n] + x_2[n]) &= \frac{3(x_1[n-1] + x_2[n-1]) + (x_1[n-2] + x_2[n-2])}{x_1[n-3] + x_2[n-3]} = \\ &= \frac{3x_1[n-1] + x_1[n-2] + 3x_2[n-1] + x_2[n-2]}{x_1[n-3] + x_2[n-3]} \neq \mathcal{H}(x_1[n]) + \mathcal{H}(x_2[n]) \quad \text{✗}\end{aligned}$$

BIBO Stability If $x[n-3] \rightarrow 0$ the output becomes unbounded. ✗

	Causal	Memory-less	Delayed	Time invariant	Linear	Stable
Summary	✓	✗	✗	✓	✗	✗

6. $y[n] = x[n]e^{n/14}$

Causal: Depends only on $x[n]$. ✓

Memory-less: Uses present input. ✓

Delay: Not a shift. ✗

Time Invariance: Coefficient depends on n . ✗

Linearity

1. Homogeneity

$$\mathcal{H}(ax[n]) = ax[n]e^{n/14} = a(x[n]e^{n/14}) = a\mathcal{H}(x[n]) \quad \checkmark$$

2. Additivity: Let the input be $x[n] = x_1[n] + x_2[n]$

$$\mathcal{H}(x_1[n] + x_2[n]) = (x_1[n] + x_2[n])e^{n/14} = x_1[n]e^{n/14} + x_2[n]e^{n/14} = \mathcal{H}(x_1[n]) + \mathcal{H}(x_2[n]) \quad \checkmark$$

BIBO Stability If $|x[n]| \leq M$, then

$$|y[n]| = |x[n]|e^{n/14} \leq Me^{n/14} \xrightarrow{n \rightarrow \infty} \infty \quad \times$$

Summary

Causal	Memory-less	Delayed	Time invariant	Linear	Stable
$\checkmark$	$\checkmark$	$\times$	$\times$	$\checkmark$	$\times$

$$\textcircled{7.} \quad y[n] = x[n]u[n]$$

$$\text{unit step sequence: } u[n] = \begin{cases} 1, & \text{if } n \geq 0 \\ 0, & \text{if } n < 0 \end{cases}$$

The system becomes

$$y[n] = \begin{cases} x[n], & \text{if } n \geq 0 \\ 0, & \text{if } n < 0 \end{cases}$$

Causal: Depends on present input. $\checkmark$ **Memory-less:** Uses only $x[n]$. $\checkmark$ **Delay:** Not a delayed input. $\times$ **Time Invariance:** Step function depends on absolute time. $\times$ **Linearity**

1. Homogeneity

$$\mathcal{H}(ax[n]) = ax[n]u[n] = a(x[n]u[n]) = a\mathcal{H}(x[n]) \quad \checkmark$$

2. Additivity: Let the input be $x[n] = x_1[n] + x_2[n]$

$$\mathcal{H}(x_1[n] + x_2[n]) = (x_1[n] + x_2[n])u[n] = x_1[n]u[n] + x_2[n]u[n] = \mathcal{H}(x_1[n]) + \mathcal{H}(x_2[n]) \quad \checkmark$$

BIBO Stability: If $|x[n]| \leq M$ then $|y[n]| \leq M$. $\checkmark$ **Summary**

Causal	Memory-less	Delayed	Time invariant	Linear	Stable
$\checkmark$	$\checkmark$	$\times$	$\times$	$\checkmark$	$\checkmark$

$$\textcircled{8.} \quad y[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[k - n + 2]$$

$$\text{unit sample sequence: } \delta[n] = \begin{cases} 1, & \text{if } n = 0 \\ 0, & \text{if } n \neq 0 \end{cases}$$

Using the sifting property: $\delta[k - n + 2] = \delta[k - (n - 2)]$ thus

$$y[n] = x[n - 2]$$

Causal: Depends on past input. ✓

Memory-less: Depends on $x[n - 2]$. ✗

Delay: Pure delay. ✓

Time Invariance: Shift property satisfied. ✓

Linearity

1. Homogeneity

$$\mathcal{H}(ax[n]) = ax[n - 2] = a\mathcal{H}(x[n]) \quad \checkmark$$

2. Additivity: Let the input be $x[n] = x_1[n] + x_2[n]$

$$\mathcal{H}(x_1[n] + x_2[n]) = (x_1[n] + x_2[n])[n - 2] = x_1[n - 2] + x_2[n - 2] = \mathcal{H}(x_1[n]) + \mathcal{H}(x_2[n]) \quad \checkmark$$

BIBO Stability If $|x[n]| \leq M$ then $|y[n]| = |x[n - 2]| \leq M \quad \checkmark$

Summary	Causal	Memory-less	Delayed	Time invariant	Linear	Stable
	✓	✗	✓	✓	✓	✓